AMAZON E-COMMERCE DATABASE

PRODUCT REVIEW AND RATING MANAGEMENT

1. Objective

The objective of this task is to create Review and Rating tables in the Amazon E-Commerce Database, insert customer reviews and product ratings, retrieve review details, calculate average product ratings, identify highly rated products, and generate a product rating report.

2. Review Table

The REVIEW table stores the reviews given by customers for products.

Column Name	Data Type	Constraint
REVIEW_ID	NUMBER	PRIMARY KEY
CUSTOMER_ID	NUMBER	NOT NULL
PRODUCT_ID	NUMBER	NOT NULL
REVIEW_TEXT	VARCHAR2(500)	NOT NULL
REVIEW_DATE	DATE	NOT NULL

```
SQL> CREATE TABLE REVIEW (
  2       REVIEW_ID NUMBER PRIMARY KEY,
  3       CUSTOMER_ID NUMBER NOT NULL,
  4       PRODUCT_ID NUMBER NOT NULL,
  5       REVIEW_TEXT VARCHAR2(500) NOT NULL,
  6       REVIEW_DATE DATE NOT NULL
  7  );

Table created.
```

3. Review Data Insertion

The following reviews were inserted into the REVIEW table.

```
SQL> INSERT INTO REVIEW VALUES
  2  (501,101,101,'Very good product and quality is excellent',
  3  TO_DATE('20-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  (502,102,102,'Good product and worth the price',
  3  TO_DATE('21-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  (503,103,103,'Product quality is average',
  3  TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  (504,104,101,'Really useful product. Loved it',
  3  TO_DATE('23-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  (505,105,104,'Amazing product and fast delivery',
  3  TO_DATE('24-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO REVIEW VALUES
  2  (506,106,102,'Product is good',
  3  TO_DATE('25-09-2026','DD-MM-YYYY'));

1 row created.
```

4. Rating Table

The RATING table stores the ratings given by customers for products.

Column Name	Data Type	Constraint
RATING_ID	NUMBER	PRIMARY KEY
CUSTOMER_ID	NUMBER	NOT NULL
PRODUCT_ID	NUMBER	NOT NULL
RATING	NUMBER(1)	NOT NULL, CHECK (1–5)
RATING_DATE	DATE	NOT NULL

```
SQL> CREATE TABLE RATING (
  2       RATING_ID NUMBER PRIMARY KEY,
  3       CUSTOMER_ID NUMBER NOT NULL,
  4       PRODUCT_ID NUMBER NOT NULL,
  5       RATING NUMBER(1) NOT NULL CHECK (RATING BETWEEN 1 AND 5),
  6       RATING_DATE DATE NOT NULL
  7  );

Table created.
```

5.Rating Data Insertion

```
SQL> INSERT INTO RATING VALUES
  2  (601,101,101,5,TO_DATE('20-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO RATING VALUES
  2  (602,102,102,4,TO_DATE('21-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO RATING VALUES
  2  (603,103,103,3,TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO RATING VALUES
  2  (604,104,101,5,TO_DATE('23-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO RATING VALUES
  2  (605,105,104,5,TO_DATE('24-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO RATING VALUES
  2  (606,106,102,4,TO_DATE('25-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO RATING VALUES
  2  (607,107,105,5,TO_DATE('26-09-2026','DD-MM-YYYY'));

1 row created.
```

6. Retrieve Product Review Details

The following query retrieves all review details and displays them according to the review date.

```
SQL> SELECT REVIEW_ID,
  2         CUSTOMER_ID,
  3         PRODUCT_ID,
  4         REVIEW_TEXT,
  5         REVIEW_DATE
  6  FROM REVIEW
  7  ORDER BY REVIEW_DATE;

 REVIEW_ID CUSTOMER_ID PRODUCT_ID
---------- ----------- ----------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       501         101        101
Very good product and quality is excellent
20-SEP-26

       502         102        102
Good product and worth the price
21-SEP-26

 REVIEW_ID CUSTOMER_ID PRODUCT_ID
---------- ----------- ----------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
```

7. Calculate Average Product Ratings

The following query calculates the total number of ratings and average rating for each product.

```
SQL> SELECT PRODUCT_ID,
  2         COUNT(RATING) AS TOTAL_RATINGS,
  3         ROUND(AVG(RATING),2) AS AVERAGE_RATING
  4  FROM RATING
  5  GROUP BY PRODUCT_ID
  6  ORDER BY AVERAGE_RATING DESC;

PRODUCT_ID TOTAL_RATINGS AVERAGE_RATING
---------- ------------- --------------
       101             2              5
       105             1              5
       104             1              5
       102             2              4
       103             1              3
```

8. Identify Highly Rated Products

Products having an average rating of 4 or above are identified using the HAVING clause.

```
SQL> SELECT PRODUCT_ID,
  2         ROUND(AVG(RATING),2) AS AVERAGE_RATING
  3  FROM RATING
  4  GROUP BY PRODUCT_ID
  5  HAVING AVG(RATING) >= 4
  6  ORDER BY AVERAGE_RATING DESC;

PRODUCT_ID AVERAGE_RATING
---------- --------------
       101              5
       105              5
       104              5
       102              4
```

9. Product Rating Report

The following query combines the PRODUCT and RATING tables to display the product name, total ratings, and average rating.

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         COUNT(R.RATING) AS TOTAL_RATINGS,
  4         ROUND(AVG(R.RATING),2) AS AVERAGE_RATING
  5  FROM PRODUCT P
  6  INNER JOIN RATING R
  7  ON P.PRODUCT_ID = R.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY AVERAGE_RATING DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_RATINGS AVERAGE_RATING
------------- --------------
       101
Laptop
            2              5

       105
T-Shirt
            1              5

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_RATINGS AVERAGE_RATING
------------- --------------

       104
Smart Watch
            1              5

       102
Smartphone

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_RATINGS AVERAGE_RATING
------------- --------------
            2              4

       103
Headphones
            1              3
```